



Dr. D. Y. Patil Educational Federation's
Dr. D. Y. Patil College of Engineering & Innovation
Approved by AICTE, Recognized by Govt. of Maharashtra,
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
Accredited by NAAC with "A" Grade



Department of Computer Engineering

Program name :- Artificial Intelligence and Machine Learning **Class:** S.Y. B.Tech **Div:-** A

Name of Student:-Kuljeet Tamchikar **Batch:-**S.Y. BTech **Course Name:-** DF

PRN No :-251112046 **Roll No:-** 32146 **Sem :-** 3rd **Course :-** BSFI (Online)



SMARTBRIDGE | SKILLWALLET

LoanWise AI: Loan Eligibility Checker & Financial Planning Platform

Project Description :-

LoanWise AI is a web-based financial planning platform built for the Indian BFSI (Banking, Financial Services and Insurance) ecosystem. It helps users understand whether they are likely to be eligible for a loan, how healthy their credit profile is, what a loan will cost them every month, and what practical steps can improve their financial position before they apply to a lender.

The platform brings four tools into one transparent workspace: a Loan Eligibility Checker, a Credit Score Analyzer, an EMI Calculator and an AI Financial Tips assistant, along with a My Records section for saved results. Users share only basic numbers such as income, obligations and loan requirements. The system never asks for PAN, Aadhaar, OTP or bank credentials.

The application provides a responsive dark-glassmorphism interface with a landing-page "LoanWise signal" summary (income fit, credit health, repayment room and next best action), factor bars for credit score drivers, amortization schedules for loans and a chat interface for AI guidance. All outputs are educational estimates and are clearly labelled as not an offer, approval, official bureau score or financial advice.

LoanWise AI is designed to be secure, accessible and easy to deploy. It uses HTML/CSS/JavaScript for the frontend, a Node.js server for API routes, the Anthropic Claude API for AI guidance, and Google Sheets-compatible storage along with browser localStorage to keep user records.



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Technology & AI :-

Frontend: HTML / CSS / JavaScript (static site served from *public/*), responsive design with reduced-motion support

Backend: Node.js (*server.js*) acting as a local and server-capable API proxy

Business Logic: Pure JavaScript module (*calculations.js*) for FOIR/DTI, EMI, credit scoring and validation, covered by unit tests

AI: Anthropic Claude API (default model *claude-sonnet-4-5*) through the server-side */api/claude* route, with a deterministic mock mode

Storage: Browser *localStorage* with a demo-memory server store; Google Sheets via Apps Script web app (optional)

API: Server-side proxy routes for Claude and Google Sheets-compatible record writes

Security: Environment variables for API keys, CORS origin control, per-minute rate limiting and AI safety guardrails

Deployment: Manus-hosted live site, Dockerfile and Netlify configuration (*netlify.toml*) for Netlify/Vercel handoff

Testing & Tooling: *pnpm test* and *pnpm run check*

Version Control: Git / GitHub

Key Features :-

- Loan eligibility estimate with FOIR/DTI, indicative score, rate band and maximum loan amount
- Clear reasons and improvement suggestions behind every eligibility result
- Weighted estimated credit score with factor bars and risk level
- Credit improvement insights and a focused 30/60/90-day plan
- Live EMI calculator with total interest and total repayment
- Amortization schedule and prepayment savings analysis
- EMI compare mode to evaluate different amounts and tenures
- AI financial tips chat with safe boundaries and consent-aware context
- Typing and error states with deterministic mock mode when no API key is present
- My Records with search, filter, CSV export and delete
- *localStorage* fallback for saved records
- Indian number formatting (e.g. 12,50,000) and 10-digit mobile number validation
- Server-side Claude and Google Sheets proxy; no API keys in frontend code
- Responsive layout for mobile, tablet and desktop
- Accessible forms and reduced-motion support
- Hard-reject and approval rules for loan eligibility, validated by unit tests



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Scenarios :-

Scenario 1: A user opens the Loan Eligibility Checker and enters their monthly income, existing EMIs, credit score and the loan amount they need. The system validates the inputs, calculates the FOIR/DTI ratio and returns an indicative score, rate band, maximum eligible amount and the reasons behind the result.

Scenario 2: A user with an average credit profile opens the Credit Score Analyzer. The system computes a weighted estimated score, shows factor bars and a risk level, and gives a focused 30/60/90-day improvement plan.

Scenario 3: A user planning a home or personal loan uses the EMI Calculator. They adjust amount, tenure and interest rate, view the amortization schedule, and compare how prepaying a little extra reduces total interest and loan duration.

Scenario 4: A user asks the AI Financial Tips assistant a plain-language question such as how to lower FOIR below 50%. The server-side proxy sends the question to Claude with safety guardrails and returns concise, India-aware guidance. The user then saves the result in My Records, searches or filters it later and exports it as a CSV file.

Milestones :-

Milestone 1 – Project Planning & Requirement Analysis

Define the objectives of LoanWise AI, identify target users, finalize the four core tools (eligibility, credit analyzer, EMI calculator and AI tips) and design the overall workflow and architecture diagrams.

Milestone 2 – Environment Setup & Project Structure

Set up the static frontend in *public/*, the Node.js *server.js* API proxy, package configuration, Dockerfile and environment-variable management through *.env.example*.

Milestone 3 – Loan Eligibility & Credit Analysis Logic

Implement the pure business logic for FOIR/DTI, loan hard rejects and approvals, rate bands, maximum amount, weighted credit scoring, risk levels and input validation including Indian number formatting and mobile validation.

Milestone 4 – EMI Calculator & Records Management

Build the live EMI calculator with amortization, prepayment savings and compare mode. Develop My Records with search, filter, CSV export, delete and localStorage fallback, plus the Google Sheets Apps Script integration.



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Milestone 5 – AI Financial Tips & Interface

Integrate the Anthropic Claude API through the secure `/api/claude` proxy with guardrails, rate limiting, consent-aware context and mock mode. Develop the responsive dark-glassmorphism interface across all pages.

Milestone 6 – Testing, Optimization & Deployment

Run unit tests for EMI known cases, amortization closure, prepayment savings, eligibility rules, credit bands and validation edge cases. Fix issues, configure production settings, and publish the application live.

Conclusion :-

LoanWise AI provides a clear and trustworthy way for users to understand their loan readiness and credit health before approaching a lender. By combining transparent financial calculations, an explainable credit analyzer, a flexible EMI planner and safe AI guidance, the platform turns complex lending criteria into practical and actionable decisions.

The project demonstrates how modern web technologies, a Node.js backend and a generative AI model can be combined to create a privacy-conscious FinTech solution for loan awareness, repayment planning and long-term financial health management.

Expected Outcome :-

LoanWise AI aims to help users make better-informed borrowing decisions. The platform provides automated eligibility estimates, credit analysis, EMI and amortization planning, AI-powered tips and saved records in a single application, available online at <https://loanwise-nemlf8ju.manus.space> with source code at github.com/kuljeettamchikar28-hub/AI-Loan-Eligibility-Checker.

The project will demonstrate practical expertise in FinTech, Generative AI, JavaScript, Node.js, REST APIs, prompt engineering, financial calculations, unit testing, responsive UI design, secure configuration and deployment, while providing a scalable foundation for future enhancements such as user authentication, machine-learning prediction models, PDF reports, Hindi and Marathi localization, bank-partner integrations and credit bureau API integration with explicit consent.